

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801G - SUSTAINABLE RIVER BASIN MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Sustainability is best defined as:
 - A) Maximizing economic growth
 - B) Development that meets present needs without compromising future generations
 - C) Conservation of resources only
 - D) Limiting globalization
 - 2 Ecological footprint represents:
 - A) Land and water area required to sustain consumption and absorb wastes
 - B) Total population size
 - C) Carbon emissions only
 - D) Industrial output
 - 3 The term Anthropocene refers to:
 - A) An ice age period
 - B) A geological epoch dominated by human influence
 - C) A natural climatic cycle
 - D) A river basin stage
 - 4 Water balance of a catchment is expressed as:
 - A) $\text{Rainfall} = \text{Runoff only}$
 - B) $\text{Evaporation} = \text{Rainfall}$
 - C) $\text{Runoff} = \text{Infiltration} + \text{Storage}$
 - D) $\text{Precipitation} = \text{Runoff} + \text{Evapotranspiration} \pm \text{Change in storage}$
 - 5 Integrated River Basin Management (IRBM) emphasizes:
 - A) Sector-wise isolated planning
 - B) Only flood control
 - C) Coordinated management of land, water, and related resources
 - D) Only irrigation development

- 6 Water scarcity occurs when:
 - A) Rainfall is low
 - B) Water demand exceeds available supply
 - C) Population density is high
 - D) Rivers dry up completely
- 7 Water governance refers to:
 - A) Political, social, and administrative systems managing water resources
 - B) Engineering design of dams
 - C) Water pricing only
 - D) Groundwater recharge methods
- 8 Water-related diseases are primarily caused by:
 - A) Excess rainfall
 - B) Climate change only
 - C) Poor water quality and sanitation
 - D) Industrialization
- 9 Wetlands are important because they:
 - A) Reduce land value
 - B) Reduce groundwater recharge
 - C) Increase desertification
 - D) Act as natural water filters and buffers
- 10 Improving water-use efficiency primarily means:
 - A) Achieving more output per unit of water used
 - B) Increasing water withdrawals
 - C) Reducing rainfall dependency
 - D) Increasing reservoir capacity
- 11 Transboundary river management requires:
 - A) Independent national planning
 - B) Cooperation and shared governance among riparian countries
 - C) Only engineering solutions
 - D) Military control
- 12 Sustainability assessment in river basin management must include:
 - A) Ecological criteria only
 - B) Economic criteria only
 - C) Ecological, economic, social, and institutional criteria
 - D) Hydrological criteria only

Q2. Solve the following.

- A) Define sustainability. Explain sustainability indicators and the concept of planetary boundaries. 6
- B) Discuss resource depletion, growth models, and the concept of ecological and carbon footprints. 6

Q3. Solve the following.

- A) Explain the hydrological cycle and water balance with reference to river basins. 6
- B) Discuss water availability, surplus, deficit, and causes of water scarcity and water crisis. 6

Q4. Solve Any Two of the following.

- A) Explain the Anthropocene concept and its relevance to climate change and climate variability. 6
 - B) Describe stream morphology and the impact of land use changes on nutrient cycles. 6
 - C) Explain river basin management and catchment terminology. 6
- Q5. Solve Any Two of the following.
- A) Discuss the relationship between water and society with respect to poverty and demography. 6
 - B) Explain water governance, integrity, and accountability in sustainable water management. 6
 - C) Explain pollution, water-related diseases, and source water protection measures. 6
- Q6. Solve Any Two of the following.
- A) Explain water conservation and efficiency measures for sustainable river basin management. 6
 - B) Discuss decision support systems and data management tools used in river basin sustainability. 6
 - C) Explain sustainability evaluation criteria (ecological, economic, social, and institutional). 6

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